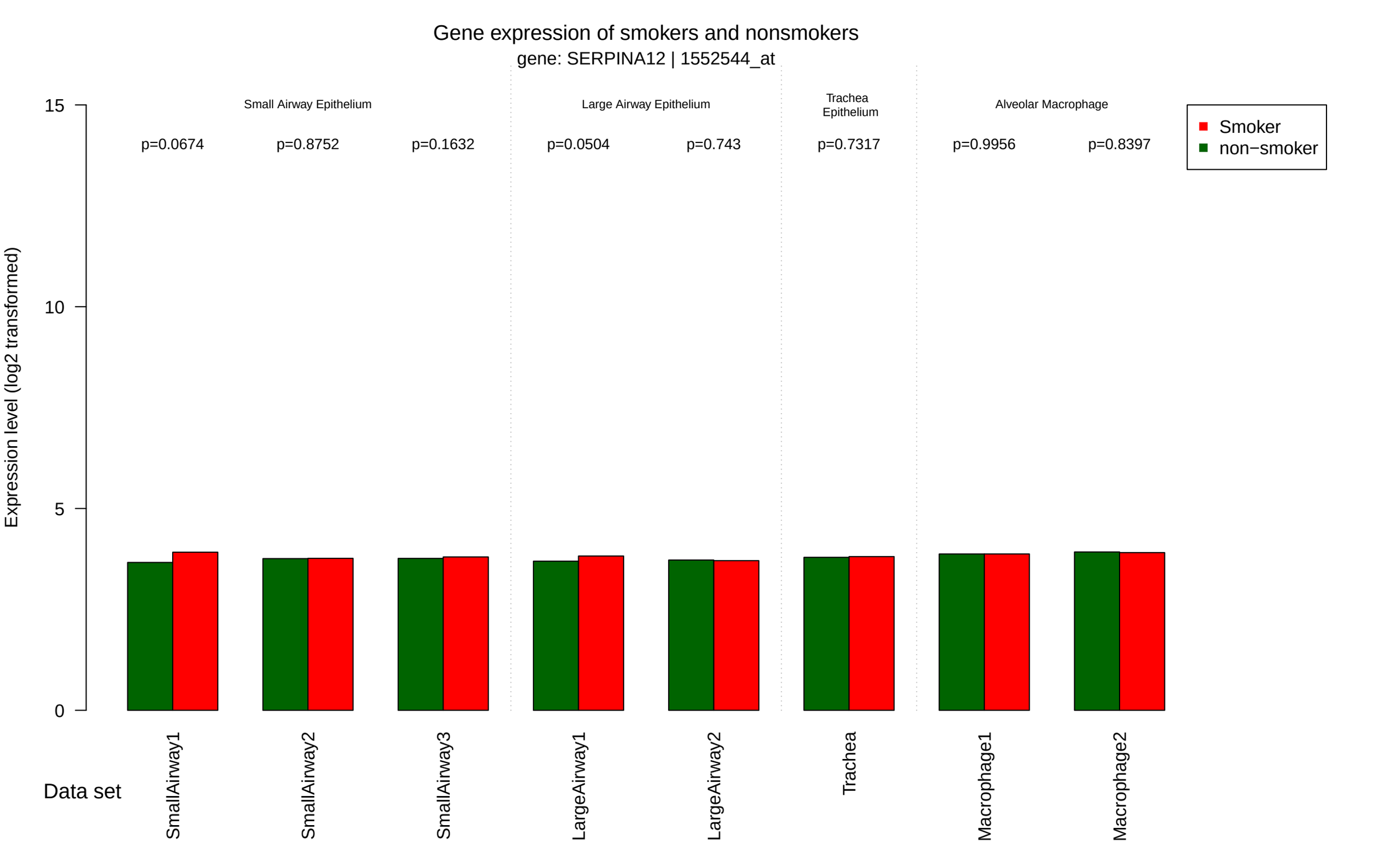
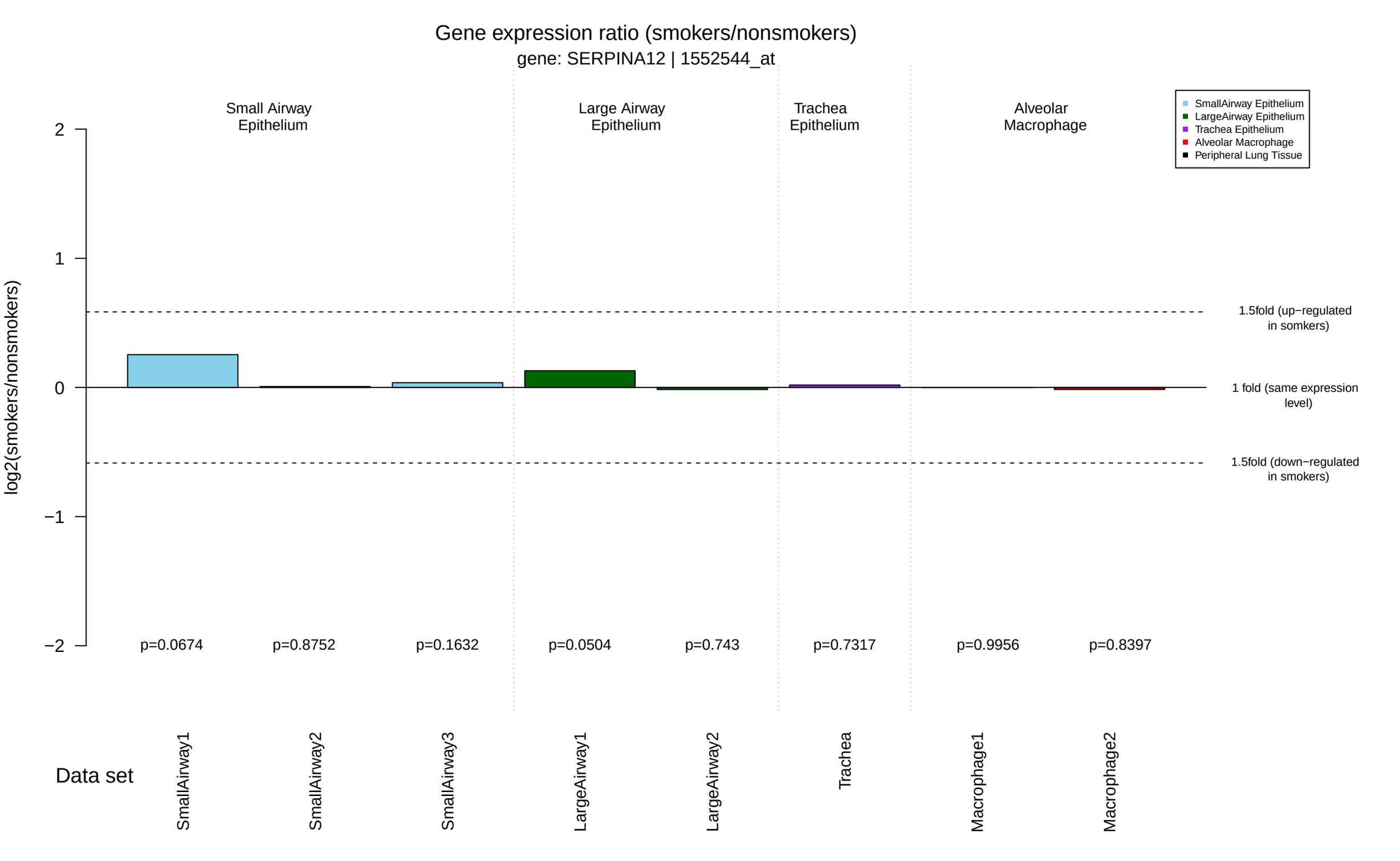
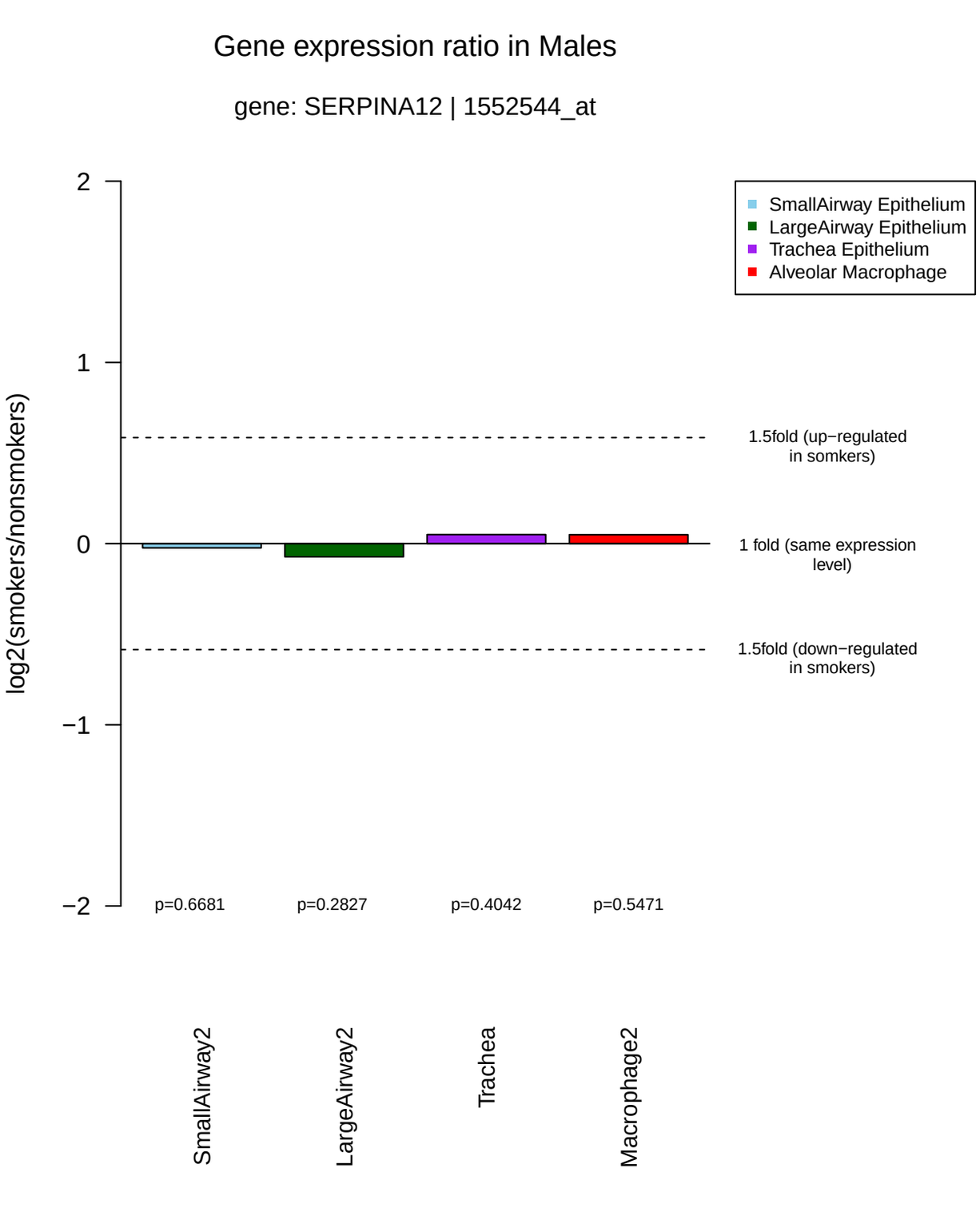
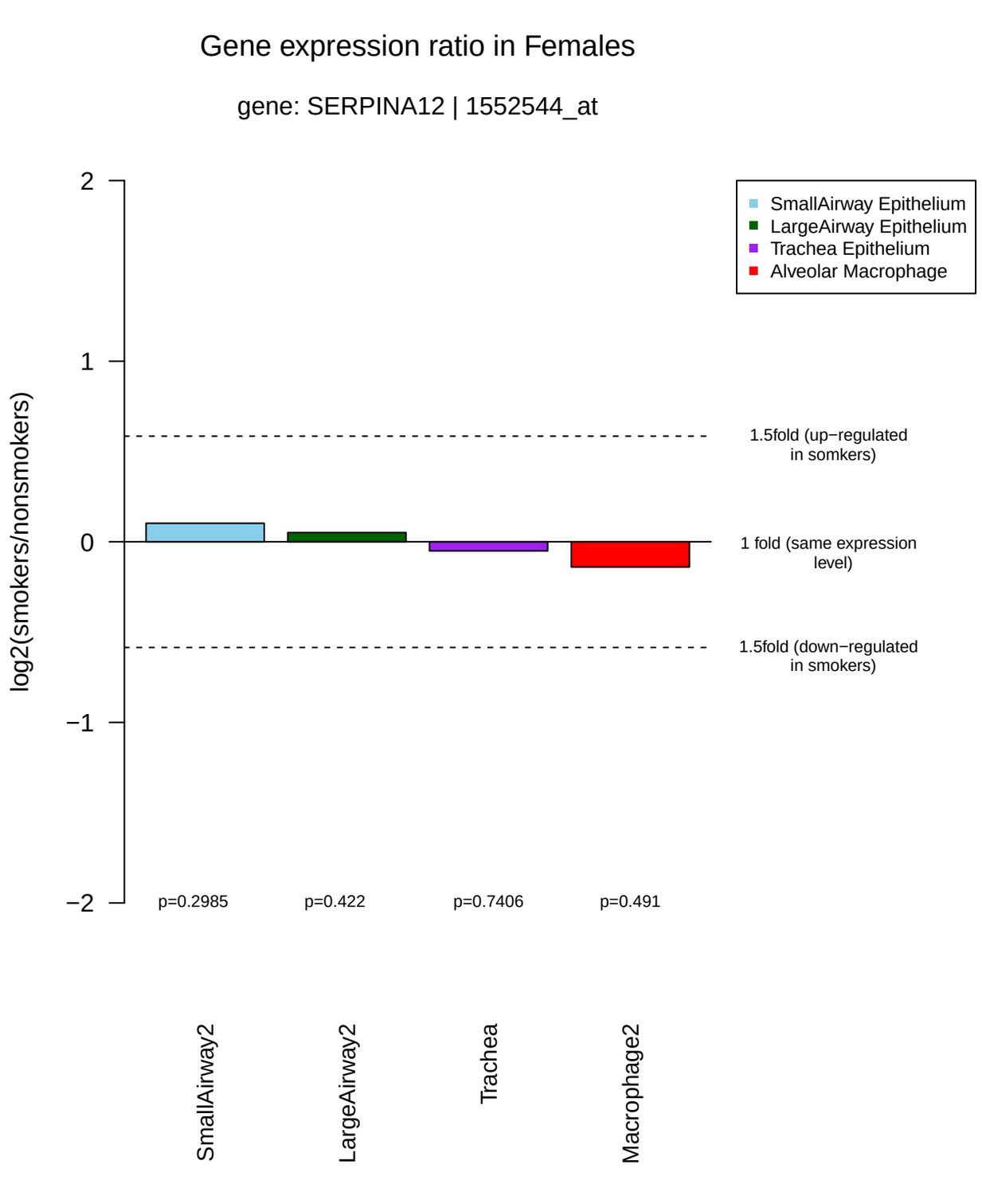
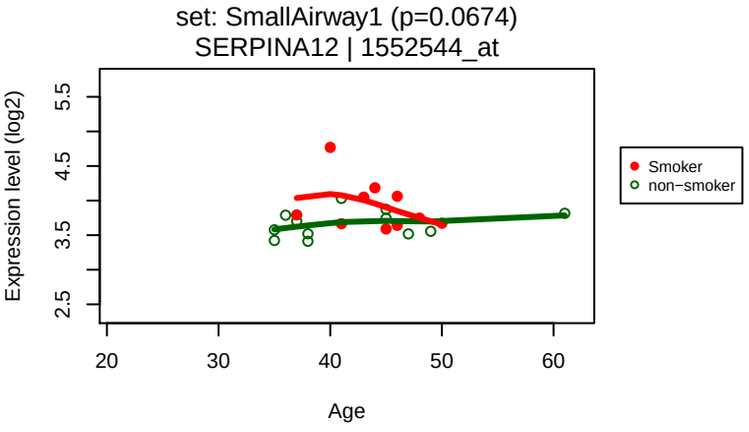
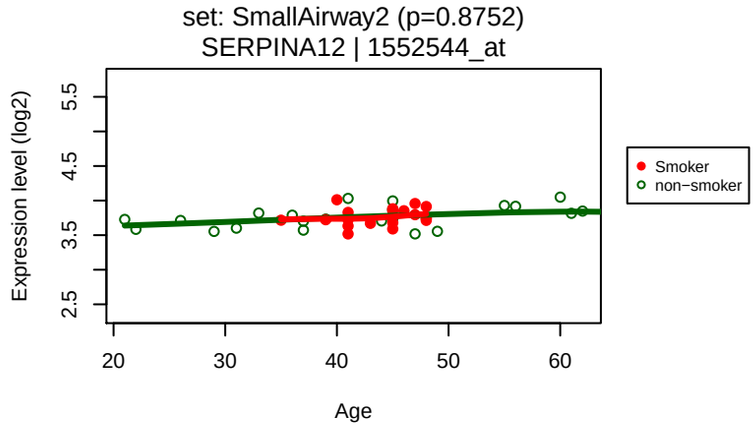

Gene expression of smokers and nonsmokers
gene: SERPINA12 | 1552544_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0674
p=0.8752
p=0.1632
p=0.0504
p=0.743
p=0.7317
p=0.9956
p=0.8397
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

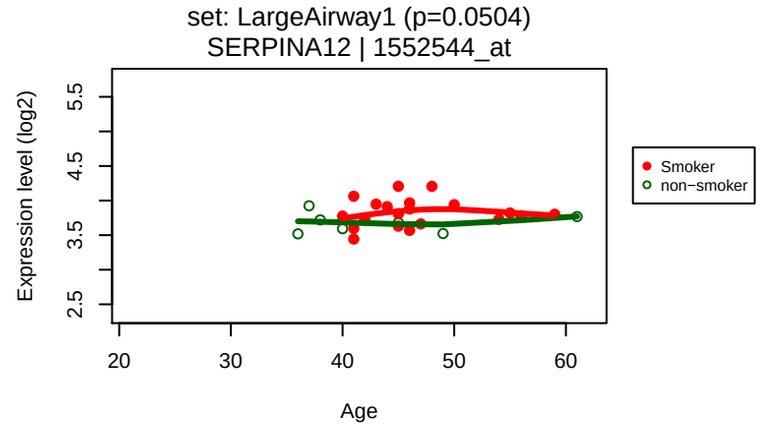

Gene expression ratio (smokers/nonsmokers)
gene: SERPINA12 | 1552544_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0674
p=0.8752
p=0.1632
p=0.0504
p=0.743
p=0.7317
p=0.9956
p=0.8397
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

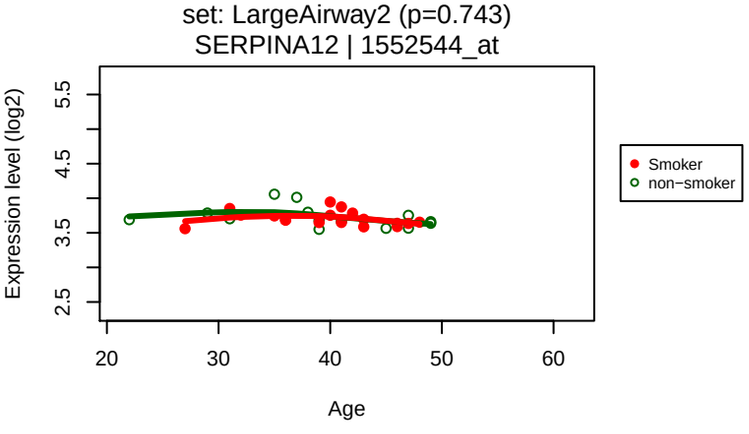

Gene expression ratio in Males
gene: SERPINA12 | 1552544_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6681
p=0.2827
p=0.4042
p=0.5471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

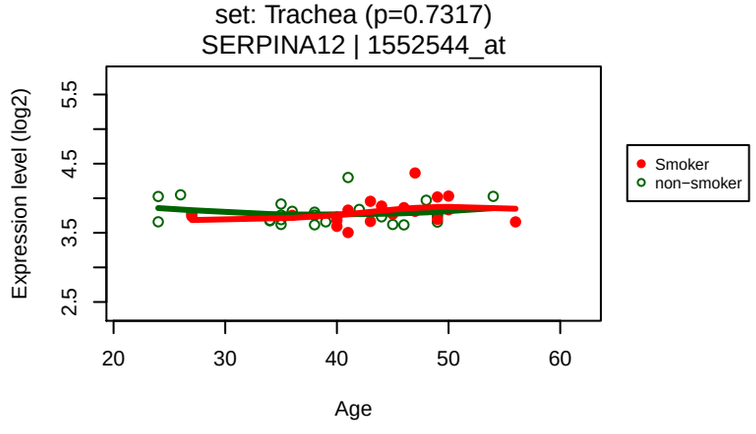

Gene expression ratio in Females
gene: SERPINA12 | 1552544_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2985
p=0.422
p=0.7406
p=0.491
SmallAirway2
LargeAirway2
Trachea
Macrophage2

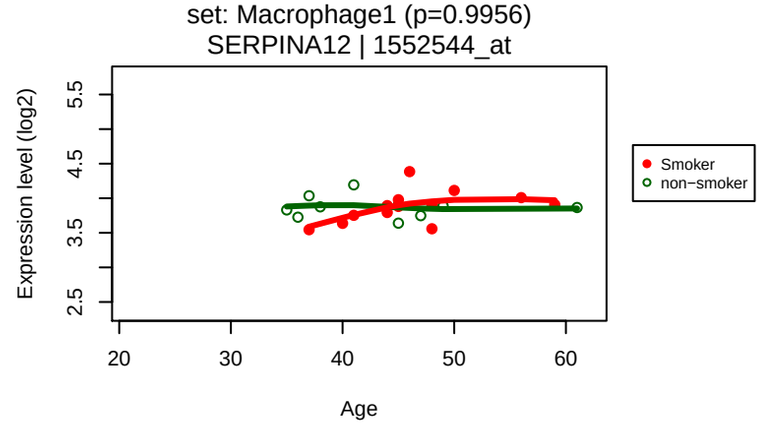

set: SmallAirway1 (p=0.0674)
SERPINA12 | 1552544_at
Expression level (log2)
Age
Smoker
non−smoker

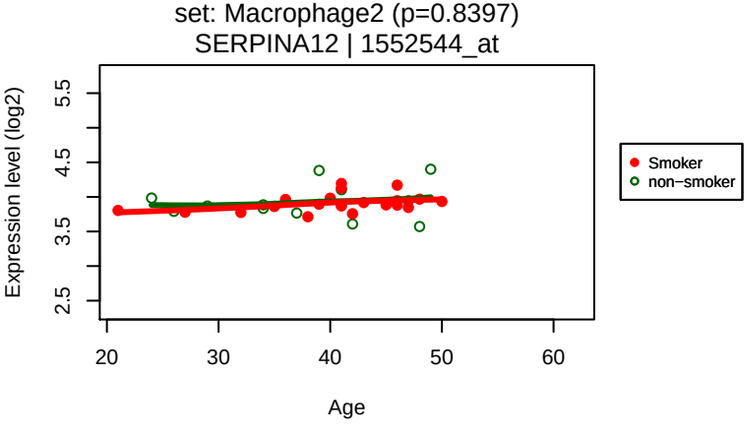

set: SmallAirway2 (p=0.8752)
SERPINA12 | 1552544_at
Expression level (log2)
Age
Smoker
non−smoker

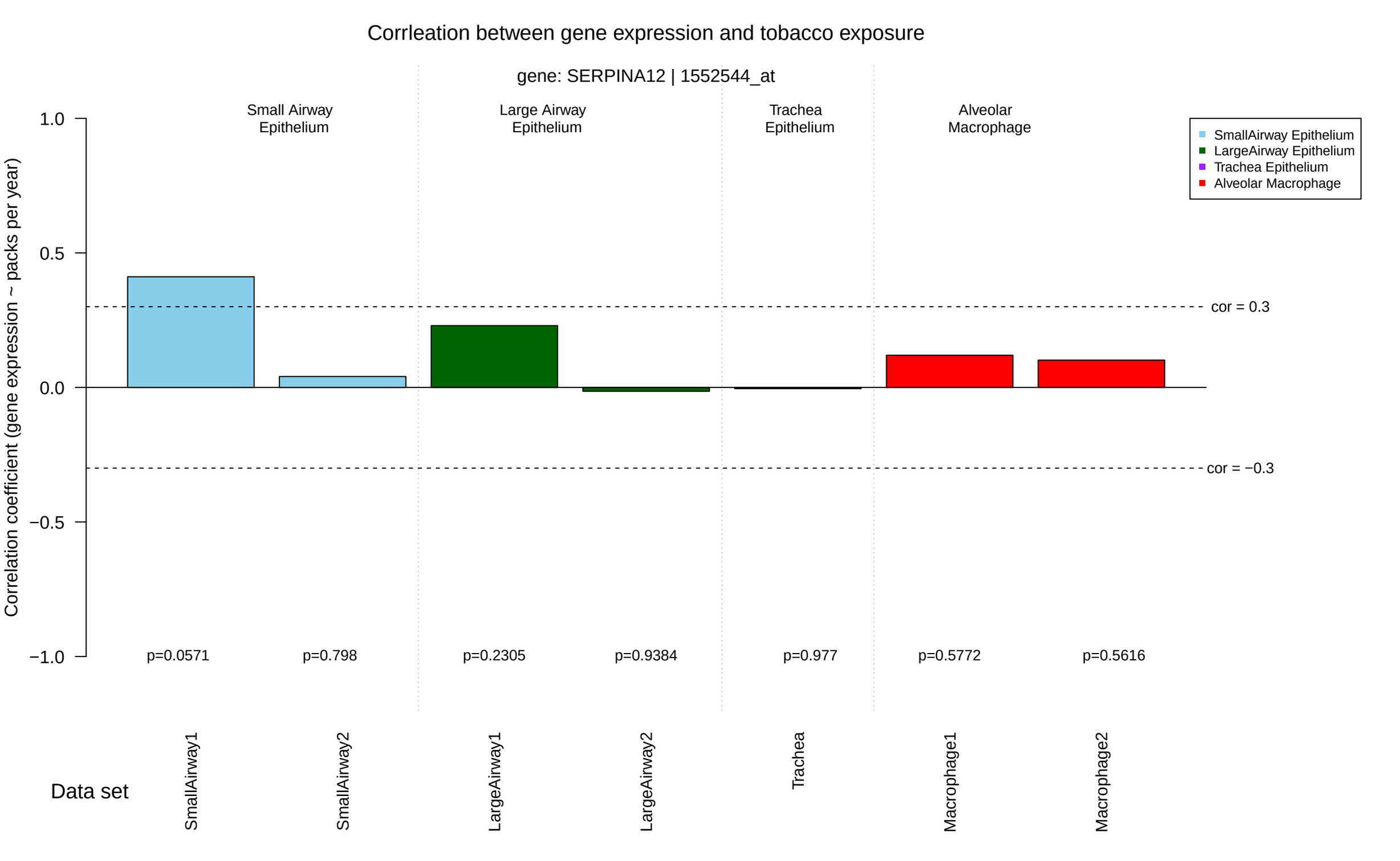

set: LargeAirway1 (p=0.0504)
SERPINA12 | 1552544_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.743)
SERPINA12 | 1552544_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7317)
SERPINA12 | 1552544_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9956)
SERPINA12 | 1552544_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8397)
SERPINA12 | 1552544_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINA12 | 1552544_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0571
p=0.798
p=0.2305
p=0.9384
p=0.977
p=0.5772
p=0.5616
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2